Jargon Buster

Bitstream: This is the coded audio data.
Codec: The component of the software that compresses and decompresses the audio file on the fly. It encodes the raw audio data into the bitstream and then decodes the bitstream while playing the file. For example, the well known codecs for MP3 are LAME and Fastenc.
Constant Bit Rate (CBR) and Variable Bit Rate (VBR): These indicate the actual bit rate at which the audio file has been encoded. CBR is a technique by which the compressed audio file is encoded at a constant bit rate throughout the song. With VBR, the bit rate of the song varies through the length of the song. For example, quieter sections are encoded at a lower bit rate and louder sections are encoded at a higher bit rate.
Digital Rights Management (DRM): This is a system used to ensure that copyrighted audio and video can be securely distributed. It involves encrypting the files with a license key that has to be present to use the file. If you lose or don't have the license key, the encrypted files cannot be played.
Spectral Band Replication (SBR): This is the extra piece of information that is written into the MP3 file as a separate stream from the normal MP3 audio data. This extra data, when read through a compatible MP3PRO decoder, allows the decoder to guess what the high frequencies sound like so they can be added to the MP3 file on the fly. This is an effective form of improving quality because the high frequencies are not stored in the MP3 file, allowing the encoder to allot bits to the more important areas of the song.
Transcoding: This is the process of converting one compressed format file to another.

While appearing similar to the MP3 standard, it improves on it by using a technology called SBR (Spectral Band Replication). Essentially, SBR reproduces those high frequency components called the PRO components that are lost during normal MP3 encoding. By combining a low bit rate MP3 file with this SBR data, we get a full-bandwidth audio file with full bass and precise treble. With SBR, MP3PRO is able to reproduce the quality of a 128 Kbps encoded MP3 file, at 64 Kbps encoding quality, resulting in files half the size of an MP3 file. However, plain MP3 files encoded at 192 KHz and above have superior sound quality compared to MP3PROs at 96 KHz—the maximum encoding quality possible within MP3PRO.

This format is backward compatible, so older portable MP3 players can read MP3PRO files simply by ignoring the PRO component. Software support on the decoder end is freely available—Winamp has introduced a plugin and recent versions of other jukebox players have also included support. Unfortunately, on the encoding front, Thompson has a demo version called the MP3PRO Audio Player that only allows 64 Kbps encoding quality. Hence, you will be required to buy software that comes with the MP3PRO codec to encode files at 80 or 96 Kbps, such as that shipped with Nero Burning Rom.

MP3PRO is aimed at applications involving streaming audio, Web casting and Internet radio. For the desktop user, this format is suitable only if the file size is just as important as audio quality—if you enjoy the quality of MP3 music at 128 Kbps, then consider shifting over to MP3PRO to save on desktop real estate.

dBpowerAMP Music, Converter 9.0 and dMC, plugins for encoding to WMA 9, Ogg Vorbis, AAC, MP3PRO decoder, dBpowerAMP Power Pack, and, Thompson MP3PRO Player/ Encoder, PsyTEL AACENC v2.15 AAC encoder and decoder (includes plug-in for Winamp), Winamp 2.81, MP3PRO decoder plugin for Winamp, WMA 9 system codecs
Find them on the Mindware CD

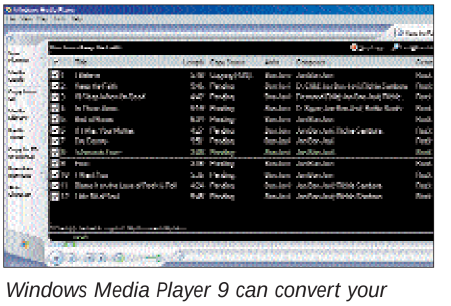
Windows Media Audio

Web site: *http://www.microsoft.com/windows/windowsmedia/default.asp*

Arguably the most popular patented and proprietary format after MP3, Windows Media Audio (WMA) proves a close rival to the MP3 standard. Made only for Windows users, this format just got better with the recent release of the WMA 9 codecs. WMA 9 can capture audio feed with a very impressive 24 bit/96 KHz sampling rate in either stereo or 5.1-channel or 7.1-channel surround sound—so you can record your music in discrete digital surround sound. Unlike the earlier versions, WMA 9 also has a lossless codec and it sounds just as good with a good old stereo arrangement. Users claim that WMA 9, encoded at a bit rate of just 48 Kbps, sounds as good as MP3 encoded at 128 Kbps. With WMAs encoded at 96 Kbps, you have sound fidelity and clarity only achievable in MP3s encoded at 192 KHz and above!

WMA supports VBR encoding, which is ideal for squeezing in maximum quality within a minimal file size. Of course, there is a caveat—Microsoft's continued support for DRM (Digital Rights Management) means that distributing copyrighted WMAs is severely restricted. The licensed file is encrypted with the license key that restricts you from storing or playing multiple copies of the file. Microsoft tracks the transfer of the license across computers too.

The hardware and software support for WMA 9 is just as good as it's for the MP3 standard. The format is backward compatible and decoder support comes in the form of the old Winamp WMA codec.



Windows Media Player 9 can convert your music CDs into high-quality WMAs

As far as encoding is concerned, you will need the new WMA 9 system codecs along with encoding software, or have Windows Media Player 9 installed.

With Microsoft's muscle backing WMA 9, it's well on its way to becoming the dominant format in the digital music arena. The format is well featured to suit diverse users, from artists wanting to distribute their music online securely to home users who need to encode a stack of CDs.

Ogg Vorbis

Web site: *http://www.xiph.org/ogg/vorbis/index.html*

This is a completely open, free source audio format that strives to replace all proprietary, patented formats. It achieved enormous popularity almost immediately after the Fraunhofer Institute decided to get tough on the MP3 standard patents